

SEC P vs NP log 2026-05-16

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-16 20:13 UTC

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1 cycles recorded

Layer-Commutator Frobenius Discrepancy Separates Read-Twice BP for IP₂

- **Verdict:** INCONCLUSIVE
- **Bridge:** Non-commutative operator discrepancy (Frobenius norm of layer-transition commutators, viewed as a Σ^b_1 -witnessing measure in bounded arithmetic à la Cook-Krajíček) $\times$ Read-twice branching programs computing the inner-product-mod-2 function IP₂
- **Recorded:** 2026-05-16 20:13 UTC
- **Entry ID:** 0e06b5f9fc52

Statement

Let P be a read-twice branching program with state set S ($|S|=s$) on n Boolean variables. For each of the $2n$ read-positions $r \in \{1, \dots, 2n\}$ let $A_r \in \mathbb{R}^{s \times s}$ be the averaged transition matrix $A_r := (T_r^0 + T_r^1)/2$, where T_r^b is the 0/1 matrix sending state u to state v if reading bit b at position r drives $u \rightarrow v$. Define the layer-commutator Frobenius discrepancy $\rho(P) := \log_2(1 + \sum_{1 \leq r < q \leq 2n} \|A_r A_q - A_q A_r\|_F^2 / s^2)$. Conjecture: (a) $\rho(P) \leq 6 \cdot \log_2(s) + 4$ for every read-twice BP P ; (b) for every read-twice BP P that computes IP₂ on $n \geq 4$ inputs, $\rho(P) \geq (1/8) \cdot n$. A single P violating either inequality refutes the conjecture; together they would force $\text{size}(P) \geq 2^{\{n/48\}}$ for read-twice BPs computing IP₂.

Rationale

Bounded-arithmetic witnessing of BP membership relies on local invariants that are Σ^b_1 -definable from the BP layer data; commutator energies of averaged layer matrices are exactly that kind of computable witness, while operator non-commutativity is the algebraic signature of the bilinear structure $\text{IP}_2(x,y) = \langle x,y \rangle_2$. Read-once BPs decompose into block-triangular layers where averaged transitions essentially commute on each reachable mass, but IP₂ forces every read-twice realization to mix x -reads and y -reads non-trivially. Because ρ is a Frobenius (L^2 spectral) quantity rather than a polynomial in the entries, the construction does not algebrize and the invariant is computed on BP structure (not the truth table), so it dodges Natural Proofs.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [0710.5926v2]

Mod 2 cohomology of 2-local finite groups of low rank - [2603.27466v1] English translation of Frobenius' and Stickelberger's "On the theory of elliptic functions" - [0906.1662v2] Frobenius morphisms of noncommutative blowups

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 211, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 148, in run_trial
    brs_bp = build_brs_bp(n)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 67, in build_brs_bp
    T0[u][v] = 1
    ~~~~~^^^
```

IndexError: list assignment index out of range

Judge reasoning

The test crashed with an IndexError in build_brs_bp before any trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the indexing bug in build_brs_bp (state-set sizing for the BRS construction) and rerun the 150-trial sweep.